Math for EE

CBSE Class 12 2015

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1. FORMULAE

Vectors and three-dimensional geometry: For vectors $\vec{a}, \vec{b} \in \mathbb{R}^3$,

1.1.5

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3, \quad \vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0. \tag{1.1.5.1}$$

1.1.1

$$\vec{a} \times \vec{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}. \tag{1.1.1.1}$$

$$\text{unit vector along } \vec{v} = \frac{\vec{v}}{\|\vec{v}\|}, \qquad \|c\vec{v}\| = |c| \, \|\vec{v}\| \, . \tag{1.1.1.2}$$

The vector product is normal to both vectors in the product. Distances and line/plane relations are The cross product is perpendicular to both input vectors: **1.1.5**

$$\vec{a} \times \vec{b} \perp \vec{a}, \qquad \vec{a} \times \vec{b} \perp \vec{b}. \tag{1.1.5.2}$$

1.1.6

$$d(\text{parallel lines}) = \frac{\left\| (\vec{b} - \vec{a}) \times \vec{v} \right\|}{\|\vec{v}\|}. \tag{1.1.6.1}$$

1.1.2

$$d(\text{skew lines}) = \frac{|(\vec{b} - \vec{a}) \cdot (\vec{v}_1 \times \vec{v}_2)|}{\|\vec{v}_1 \times \vec{v}_2\|}. \tag{1.1.2.2}$$

1.1.10

$$\text{line parallel to plane} \iff \vec{d} \cdot \vec{n} = 0. \tag{1.1.10.1}$$

$$\text{plane through } \vec{r}_0 \text{ with normal } \vec{n}: \quad \vec{n} \cdot (\vec{r} - \vec{r}_0) = 0. \tag{1.1.10.2}$$

1.1.15

$$\text{line with point } \vec{A} \text{ and direction } \vec{d}: \quad \vec{r} = \vec{A} + t\vec{d}. \tag{1.1.15.1}$$

Matrices and determinants:

1.1.28

$$\det(AB) = \det(A) \det(B), \qquad \det(A^n) = (\det A)^n. \tag{1.1.28.1}$$

1.1.30

$$A^{-1} = \frac{\operatorname{adj}(A)}{\det A}, \qquad \det A \neq 0. \tag{1.1.30.1}$$

$$A\vec{x} = \vec{b} \quad \Longrightarrow \quad \vec{x} = A^{-1}\vec{b} \quad (A \text{ invertible}). \tag{1.1.30.2}$$

1.1.37

$$[A \mid \vec{b}] \xrightarrow{\text{elementary row operations}} [I \mid \vec{x}] \quad \text{(Gaussian elimination).} \tag{1.1.37.1}$$

The identity used to factor the determinant in Q36 is **1.1.36**

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}\big((a-b)^2 + (b-c)^2 + (c-a)^2\big). \tag{1.1.36.1}$$

Differentiation and differential equations: For $y' + P(x)y = Q(x)$, the integrating factor is **1.1.24**

$$\mu(x) = e^{\int P(x)\,dx}. \tag{1.1.24.1}$$

The logarithmic derivative is **1.1.40**

$$\frac{d}{dx}\log|u(x)| = \frac{u'(x)}{u(x)}, \qquad u(x) \neq 0. \tag{1.1.40.1}$$

The same identity gives **1.1.42**

$$\frac{d}{dx}\log|u(x)| = \frac{u'(x)}{u(x)}, \qquad u(x) \neq 0. \tag{1.1.42.1}$$

1.1.25

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \qquad \frac{d^2y}{dx^2} = \frac{d}{dt}(dy/dx)\Big/\frac{dx}{dt}. \tag{1.1.25.2}$$

1.1.20

$$y' + P(x)y = Q(x), \quad \mu(x) = e^{\int P(x)\,dx}, \quad \frac{d}{dx}(\mu y) = \mu Q. \tag{1.1.20.1}$$

1.1.22

$$y = vx \quad \Longrightarrow \quad y' = v + xv' \quad \text{(homogeneous first-order ODE).} \tag{1.1.22.1}$$

1.1.21

$$(1 + (y')^2)^3 = r^2(y'')^2 \quad \text{for a circle family of fixed radius } r. \tag{1.1.21.1}$$

1.1.25

$$\frac{d}{dx}\log|u(x)| = \frac{u'(x)}{u(x)}, \qquad u(x) \neq 0. \tag{1.1.25.1}$$

Integration and area: A useful definite-integral antiderivative is **1.1.39**

$$\int \frac{du}{a^2 - u^2} = \frac{1}{2a}\log\left|\frac{a+u}{a-u}\right| + C. \tag{1.1.39.1}$$

After polynomial division, partial fractions can be written as **1.1.43**

$$\frac{P(x)}{(x-a)(x^2+b^2)} = Q(x) + \frac{A}{x-a} + \frac{Bx+C}{x^2+b^2}. \tag{1.1.43.1}$$

The trigonometric integrand in Question 44 can be split using **1.1.44**

$$\frac{5\sin x + 3\cos x}{\sin x + \cos x} = 4 + \frac{\sin x - \cos x}{\sin x + \cos x}. \tag{1.1.44.1}$$

Integration by parts states **1.1.45**

$$\int u\,dv = uv - \int v\,du. \tag{1.1.45.2}$$

Area between curves is **1.1.47**

$$A = \int_a^b |y_{\text{upper}} - y_{\text{lower}}|\,dx. \tag{1.1.47.1}$$

1.1.38

$$\int \frac{du}{a^2 - u^2} = \frac{1}{2a}\log\left|\frac{a+u}{a-u}\right| + C. \tag{1.1.38.1}$$

$$\int \frac{du}{1+u^2} = \tan^{-1} u + C. \tag{1.1.38.2}$$

1.1.46

$$\int \frac{u\,du}{\sqrt{1+u^2}} = \sqrt{1+u^2} + C. \tag{1.1.46.2}$$

$$A = \int_a^b |y_{\text{upper}} - y_{\text{lower}}|\,dx. \tag{1.1.46.1}$$

1.1.41

$$\frac{P(x)}{(x-a)(x^2+b^2)} = Q(x) + \frac{A}{x-a} + \frac{Bx+C}{x^2+b^2} \quad \text{(partial fractions after division).} \tag{1.1.41.1}$$

1.1.45

$$\int u\,dv = uv - \int v\,du. \tag{1.1.45.1}$$

Functions, relations and operations: Continuity at an interior point requires matching one-sided limits: **1.1.52**

$$\lim_{x\to a^-} f(x) = f(a) = \lim_{x\to a^+} f(x). \tag{1.1.52.1}$$

At an interior point, continuity requires matching one-sided limits; differentiability additionally requires matching one-sided derivatives: **1.1.48**

$$\lim_{x\to a^-} f(x) = f(a) = \lim_{x\to a^+} f(x), \qquad f'_-(a) = f'_+(a) = f'(a). \tag{1.1.48.1}$$

1.1.49

$$\begin{aligned} &\text{reflexive: } (a,a) \in R, \\ &\text{symmetric: } (a,b) \in R \Rightarrow (b,a) \in R, \\ &\text{transitive: } (a,b),(b,c) \in R \Rightarrow (a,c) \in R. \end{aligned} \tag{1.1.49.1}$$

1.1.50

For componentwise addition on $\mathbb{R}^2$: $\quad e=(0,0), \quad (a,b)^{-1}=(-a,-b).$ (1.1.50.1)

1.1.51

$a*b=(a+b) \bmod n, \quad 0$ is the identity, $\quad (-a) \bmod n$ is the additive inverse. (1.1.51.1)

Probability: For a binomial random variable, **1.1.56**

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}. \tag{1.1.56.1}$$

The law of total probability is **1.1.57**

$$P(E)=\sum_i P(E \mid A_i)P(A_i). \tag{1.1.57.1}$$

Bayes' theorem is **1.1.58**

$$P(A \mid E)=\frac{P(E \mid A)P(A)}{P(E)}. \tag{1.1.58.2}$$

1.1.58

$$P(A \mid E)=\frac{P(A \cap E)}{P(E)}. \tag{1.1.58.1}$$

1.1.54

$$P(A \mid E)=\frac{P(E \mid A)P(A)}{P(E)}. \tag{1.1.54.1}$$

1.1.53

$$X \sim \mathrm{Bin}(n,p): \quad P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}. \tag{1.1.53.1}$$

$$G(z)=\sum_j P(D=j)z^j. \tag{1.1.53.2}$$

$G_5(z)=(0.4z+0.6z^{-1})^5$ for the five-step walk.

1.1.55

$$E[X]=\sum_x xP(X=x), \quad \mathrm{Var}(X)=E[X^2]-E[X]^2. \tag{1.1.55.1}$$

$$P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}. \tag{1.1.55.2}$$

1.1.54

$$P(E)=\sum_i P(E \mid A_i)P(A_i). \tag{1.1.54.2}$$

Linear programming: For a linear programme, compare the objective at every feasible vertex: **1.1.60**

$$\text{LPP optimum} = \text{best objective value among feasible vertices.} \tag{1.1.60.1}$$

1.1.59

Evaluate the objective at each feasible corner; an optimum occurs at a feasible vertex. (1.1.59.1)

Coordinate geometry and optimisation: The tangent to $x^2 + y^2 = r^2$ at $(r\cos\theta, r\sin\theta)$ is **1.1.64**

$$x\cos\theta + y\sin\theta = r. \tag{1.1.64.1}$$

For the ellipse $x^2/a^2 + y^2/b^2 = 1$, a parametrisation is $(a\cos t, b\sin t)$, and triangle area is one half base times perpendicular height: **1.1.65**

$$\text{ellipse point } (a\cos t, b\sin t), \quad A_{\triangle} = \frac{1}{2}bh. \tag{1.1.65.1}$$

Engineering methods: choosing the right tool: Matrix methods are used for vector and matrix conditions and for Gaussian elimination. The five-step random walk and the small binomial problem use probability-generating polynomial coefficients where it makes the probability structure clearer. Laplace transforms are useful for suitable linear time-domain differential equations with initial conditions; they are not forced into the nonlinear or variable-coefficient differential equations in this set where they would not simplify the solution.

2. Problems

Vectors and three-dimensional geometry:

1.2.1 Find a vector of magnitude $\sqrt{171}$ perpendicular to both $\vec{a} = \begin{bmatrix} 1 \\ -2 \\ -3 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$.

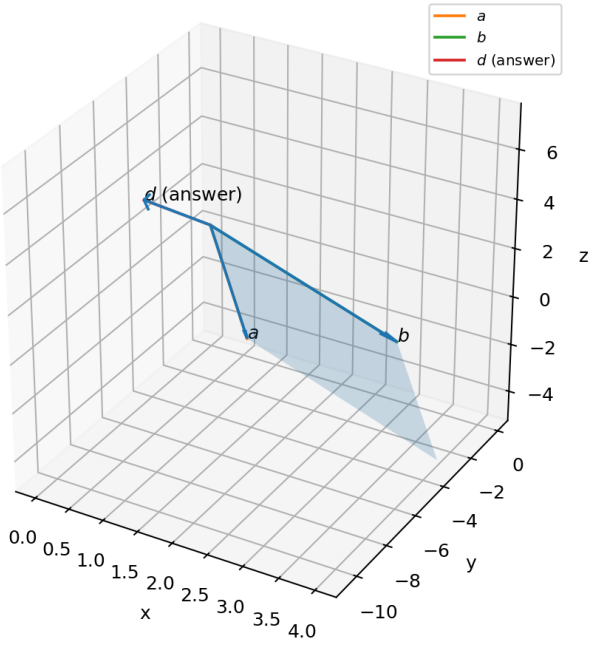
Solution. A vector perpendicular to both is parallel to their cross product:

$$\vec{a} \times \vec{b} = \begin{bmatrix} 1 \\ -7 \\ 5 \end{bmatrix}, \qquad \left\| \vec{a} \times \vec{b} \right\| = \sqrt{75} = 5\sqrt{3}. \tag{1.2.0.1}$$

Scale this direction to the required magnitude:

$$\vec{d} = \frac{\sqrt{171}}{5\sqrt{3}} \begin{bmatrix} 1 \\ -7 \\ 5 \end{bmatrix} = \pm \frac{\sqrt{57}}{5} \begin{bmatrix} 1 \\ -7 \\ 5 \end{bmatrix}.$$

Q1: the answer vector is normal to the plane spanned by a and b



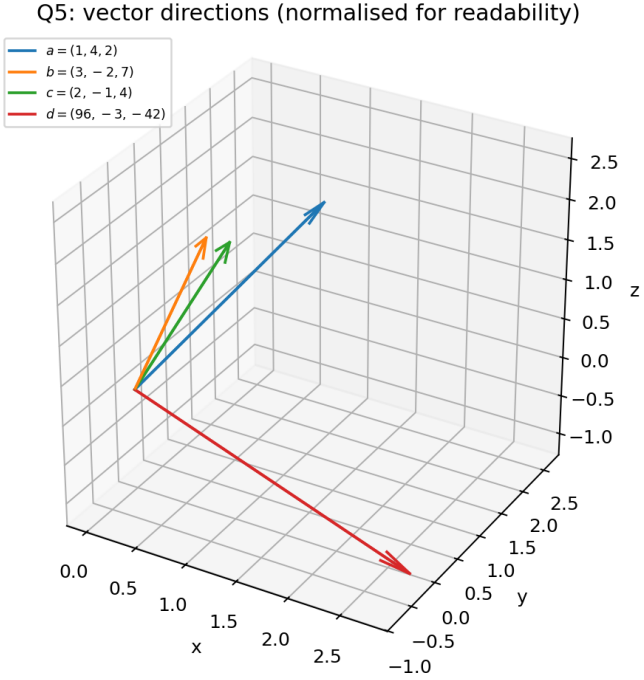
Code reference – plotting and numerical verification

`codes/Q01.py`

1.2.5 Let $\vec{a} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 3 \\ -2 \\ 7 \end{bmatrix}$ and $\vec{c} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$. Find $\vec{d}$ perpendicular to both $\vec{a}, \vec{b}$ with $\vec{c} \cdot \vec{d} = 27$.

Solution. The required direction is $\vec{a} \times \vec{b} = (32, -1, -14)^T$. Set $\vec{d} = t(32, -1, -14)^T$. Since $\vec{c} \cdot (\vec{a} \times \vec{b}) = 9$, the constraint gives $9t = 27$, hence

$$\vec{d} = \begin{bmatrix} 96 \\ -3 \\ -42 \end{bmatrix}.$$



Code reference – plotting and numerical verification

`codes/Q05.py`

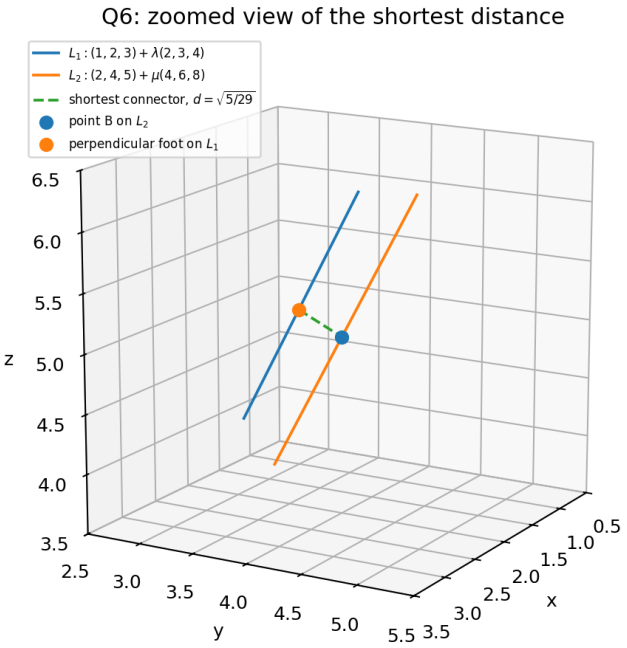
1.2.6 Find the shortest distance between $\vec{r} = (1, 2, 3) + \lambda(2, 3, 4)$ and $\vec{r} = (2, 4, 5) + \mu(4, 6, 8)$.

Solution. The direction vectors are parallel, since $(4, 6, 8) = 2(2, 3, 4)$. For parallel lines the distance is the perpendicular distance from one point to the other line:

$$d = \frac{\|(B - A) \times \vec{v}\|}{\|\vec{v}\|}, \quad B - A = (1, 2, 2), \quad \vec{v} = (2, 3, 4).$$

Here $(B - A) \times \vec{v} = (2, 0, -1)$, so

$$d = \frac{\sqrt{5}}{\sqrt{29}}.$$



Code reference – plotting and numerical verification

`codes/Q06.py`

Lines and planes:

1.2.10 Find the plane through the intersection of $2x + y - z = 3$ and $5x - 3y + 4z + 9 = 0$ that is parallel to $\frac{x-1}{2} = \frac{y-3}{-4} = \frac{5-z}{-5}$.

Solution. A plane through the intersection has equation

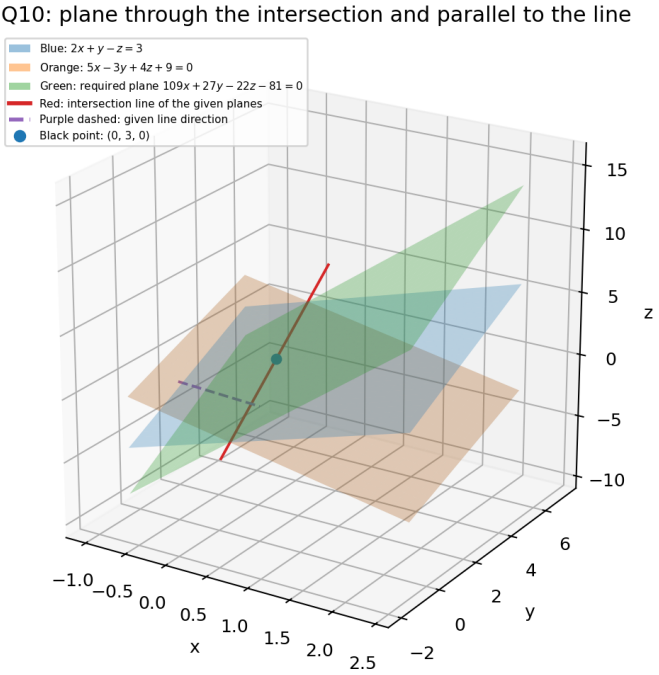
$$(2x + y - z - 3) + \lambda(5x - 3y + 4z + 9) = 0.$$

The line direction is $(2, -4, 5)$. Its dot product with the plane normal $(2+5\lambda, 1-3\lambda, -1+4\lambda)$ must vanish:

$$2(2 + 5\lambda) - 4(1 - 3\lambda) + 5(-1 + 4\lambda) = 0,$$

so $\lambda = 5/42$. Multiplying by 42 and simplifying gives

$$109x + 27y - 22z - 81 = 0.$$



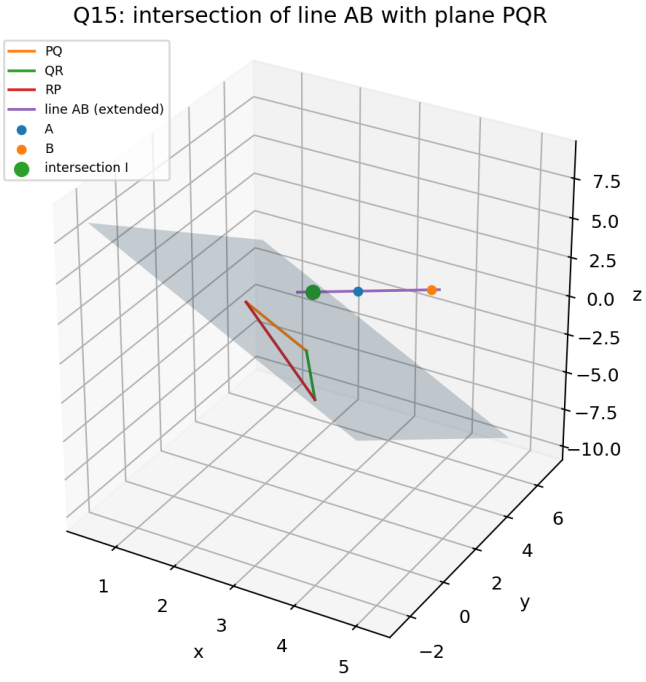
Code reference – plotting and numerical verification

`codes/Q10.py`

1.2.15 Find where the line through $A(3,4,1)$ and $B(5,1,6)$ meets the plane through $P(2,1,2)$, $Q(3,1,0)$ and $R(4,-2,1)$.

Solution. A normal to the plane is $(Q-P)\times(R-P) = (1,0,-2)\times(2,-3,-1) \propto (2,1,1)$. Thus the plane is $2x+y+z=7$. The line is $(3,4,1)+t(2,-3,5)$. Substitution into the plane yields $4+6t=0$, so $t=-2/3$. The intersection is

$$\left(\frac{5}{3}, 6, -\frac{7}{3}\right).$$



Code reference – plotting and numerical verification

`codes/Q15.py`

Differential equations:

1.2.20 Find the integrating factor of $x \log x \, \frac{dy}{dx} + y = 2 \log x$.

Solution. In standard form,

$$y' + \frac{1}{x \log x} y = \frac{2}{x}.$$

Thus the integrating factor is

$$\exp\left(\int \frac{dx}{x \log x} \right) = \exp(\log |\log x|).$$

On a domain where $\log x > 0$, this is $\log x$ (more generally the factor can be taken as $|\log x|$ on an interval not crossing $x = 1$).

1.2.21 Find the differential equation of the family $(x - h)^2 + (y - k)^2 = r^2$, where h, k are arbitrary constants and r is fixed.

Solution. Differentiate once:

$$(x - h) + (y - k) y' = 0.$$

Differentiate again: $1 + (y')^2 + (y - k) y'' = 0$, giving $y - k = -(1 + (y')^2)/y''$. Substitute

this and $x-h=-(y-k)y'$ into the original circle equation. Then

$$\frac{(1+(y')^2)^3}{(y'')^2}=r^2,\quad (1+(y')^2)^3=r^2(y'')^2.$$

1.2.22 Show that $(x-y)\frac{dy}{dx}=x+2y$ is homogeneous and solve it.

Solution. The derivative is a ratio of homogeneous first-degree expressions. Put $y=vx$, so $y'=v+xv'$. Substitution gives

$$x\frac{dv}{dx}=\frac{1+v+v^2}{1-v},\qquad \frac{1-v}{1+v+v^2}\,dv=\frac{dx}{x}.$$

Integrating,

$$-\frac{1}{2}\ln(v^2+v+1)+\sqrt{3}\tan^{-1}\!\left(\frac{2v+1}{\sqrt{3}}\right)=\ln|x|+C.$$

Using $v=y/x$ gives the implicit solution

$$-\tfrac{1}{2}\ln\!\left(\frac{y^2}{x^2}+\frac{y}{x}+1\right)+\sqrt{3}\tan^{-1}\!\left(\frac{2y/x+1}{\sqrt{3}}\right)=\ln|x|+C.$$

1.2.24 Find the particular solution of $y'+y\tan x=3x^2+x^3\tan x$, given $y=0$ at $x=\pi/3$ and $x\neq\pi/2$.

Solution. The integrating factor is $e^{\int\tan x\,dx}=\sec x$. Multiplication gives

$$\frac{d}{dx}(y\sec x)=3x^2\sec x+x^3\sec x\tan x=\frac{d}{dx}(x^3\sec x).$$

So $y\sec x=x^3\sec x+C$ and $y=x^3+C\cos x$. From $y(\pi/3)=0$, $C=-2\pi^3/27$. Therefore

$$y=x^3-\frac{2\pi^3}{27}\cos x.$$

Code reference – numerical verification

codes/Q24.py

1.2.25 If $y=\log\left(\frac{x}{a+bx}\right)^x$, prove that $x^3y''=(xy'-y)^2$.

Solution. Interpret the expression as $y=x\log\left(\frac{x}{a+bx}\right)$. Then

$$y'=\log\left(\frac{x}{a+bx}\right)+\frac{a}{a+bx},\qquad y''=\frac{a^2}{x(a+bx)^2}.$$

Thus $xy'-y=ax/(a+bx)$ and

$$x^3y''=\frac{a^2x^2}{(a+bx)^2}=(xy'-y)^2.$$

Matrices and determinants:

1.2.28 If $A=\begin{bmatrix}\cos\theta & \sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$, find $\det(A^n)$ for natural n.

Solution. Determinants are multiplicative, so $\det(A^n) = (\det A)^n$. Now $\det A = \cos^2\theta - \sin^2\theta = \cos 2\theta$. Thus

$$\det(A^n) = (\cos 2\theta)^n.$$

1.2.30 Find the inverse of $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 0 \end{bmatrix}$ using elementary row operations.

Solution. Augment the matrix by I and row-reduce to $[I \mid A^{-1}]$. The determinant is -1, so the matrix is invertible. The result is

$$A^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -9 & 6 & -2 \\ 5 & -3 & 1 \end{bmatrix}.$$

The product check $AA^{-1} = I$ is carried out in code.

Code reference – numerical verification

codes/Q30.py

1.2.36 If $a+b+c \neq 0$ and $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, prove that $a = b = c$.

Solution. Expanding and factorising the determinant gives

$$-(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 0.$$

The first factor is nonzero by assumption, so $a^2+b^2+c^2-ab-bc-ca = 0$. But

$$a^2+b^2+c^2-ab-bc-ca = \tfrac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2].$$

Each square must be zero, hence $a = b = c$.

1.2.37 A trust receives Rs. 30,000 per month. Half is spent on care, and 2% of the amount spent is collected from beneficiaries. The unspent balance is deposited in a bank. Find the monthly interest rate needed for total monthly income of Rs. 1800.

Solution. Let u be the monthly contribution collected from beneficiaries and v the bank interest. Half of Rs. $30,000$ is spent, so

$$u = 0.02(15000) = 300.$$

The total monthly income is Rs. 1800, so

$$u + v = 1800, \qquad u = 300.$$

Write this system as an augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 1800 \\ 1 & 0 & 300 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 1800 \\ 0 & -1 & -1500 \end{array}\right] \xrightarrow{R_2 \leftarrow -R_2} \left[\begin{array}{cc|c} 1 & 1 & 1800 \\ 0 & 1 & 1500 \end{array}\right].$$

Eliminate v from the first row:

$$R_1 \leftarrow R_1 - R_2: \qquad \left[\begin{array}{cc|c} 1 & 0 & 300 \\ 0 & 1 & 1500 \end{array}\right].$$

Thus $u = 300$ and $v = 1500$. The deposit is Rs. 15000, therefore

$$r = \frac{1500}{15000} \times 100\% = 10\% \text{ per month.}$$

Code reference – numerical verification

`codes/Q37.py`

Integration and area:

1.2.38 Evaluate $I = \displaystyle\int_0^{\pi/2} \frac{\cos^2 x}{1 + 3\sin^2 x}\, dx.$

Solution. Let $t = \tan x$. Then $dx = dt/(1+t^2)$ and the integral becomes

$$I = \int_0^{\infty} \frac{dt}{(1+t^2)(1+4t^2)} = \int_0^{\infty} \left[-\frac{1}{3(1+t^2)} + \frac{4}{3(1+4t^2)}\right] dt.$$

Evaluating the arctangent limits gives $I = -\pi/6 + \pi/3 = \pi/6$.

1.2.39 Evaluate $\displaystyle\int_0^{\pi/4} \frac{\sin x + \cos x}{3 + \sin 2x}\, dx.$

Solution. Set $u = \sin x - \cos x$, so $du = (\sin x + \cos x)dx$ and $\sin 2x = 1 - u^2$. The limits become $u = -1$ and $u = 0$. Hence

$$I = \int_{-1}^{0} \frac{du}{4 - u^2} = \left[\frac{1}{4} \ln\left|\frac{2+u}{2-u}\right|\right]_{-1}^{0} = \frac{1}{4}\ln 3.$$

1.2.40 Evaluate $\displaystyle\int \frac{\sin x - x\cos x}{x(x + \sin x)}\, dx.$

Solution. Since

$$\frac{d}{dx} \ln\left|\frac{x}{x+\sin x}\right| = \frac{\sin x - x\cos x}{x(x+\sin x)},$$

the integral is $\ln\left|\frac{x}{x+\sin x}\right| + C$.

1.2.41 Evaluate $\displaystyle\int \frac{x^3}{(x-1)(x^2+1)}\, dx.$

Solution. Polynomial division and partial fractions give

$$\frac{x^3}{(x-1)(x^2+1)} = 1 + \frac{1}{2(x-1)} + \frac{x-1}{2(x^2+1)}.$$

Therefore

$$x + \frac{1}{2}\ln|x-1| + \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\tan^{-1} x + C.$$

1.2.42 Find $\displaystyle\int \frac{x\, dx}{1 + x\tan x}.$

Solution. Rewrite the integrand as $x\cos x/(\cos x + x\sin x)$. Since $d(\cos x + x\sin x)/dx = x\cos x$,

$$\int \frac{x\,dx}{1 + x\tan x} = \ln|\cos x + x\sin x| + C.$$

1.2.43 Evaluate $\displaystyle\int \frac{x^4}{(x-1)(x^2+1)}\,dx$.

Solution. Division gives $x^4/[(x-1)(x^2+1)] = x + 1 + 1/[(x-1)(x^2+1)]$. Decompose the last term as

$$\frac{1}{(x-1)(x^2+1)} = \frac{1}{2(x-1)} - \frac{x+1}{2(x^2+1)}.$$

Thus

$$\frac{x^2}{2} + x + \frac{1}{2}\ln|x-1| - \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\tan^{-1}x + C.$$

1.2.44 Evaluate $\displaystyle\int_0^{\pi/2} \frac{5\sin x + 3\cos x}{\sin x + \cos x}\,dx$.

Solution. Write the numerator as $4(\sin x + \cos x) + (\sin x - \cos x)$. Then

$$I = 4\int_0^{\pi/2} dx - [\ln(\sin x + \cos x)]_0^{\pi/2} = 2\pi.$$

1.2.45 Find $\displaystyle\int \left[\log(\log x) + \frac{1}{(\log x)^2}\right] dx$.

Solution. Differentiate $x\log(\log x) - x/\log x$ to check that it equals the integrand. Hence

$$x\log(\log x) - \frac{x}{\log x} + C, \qquad x > 1 \text{ for a real-valued } \log(\log x).$$

1.2.46 If the area bounded by $y^2 = 16ax$ and $y = 4mx$ is $a^2/2$, find m.

Solution. Use the upper branch $y = 4\sqrt{ax}$. At an intersection with $y = 4mx$,

$$4\sqrt{ax} = 4mx \quad\Longrightarrow\quad x = 0 \quad \text{or} \quad x = \frac{a}{m^2}.$$

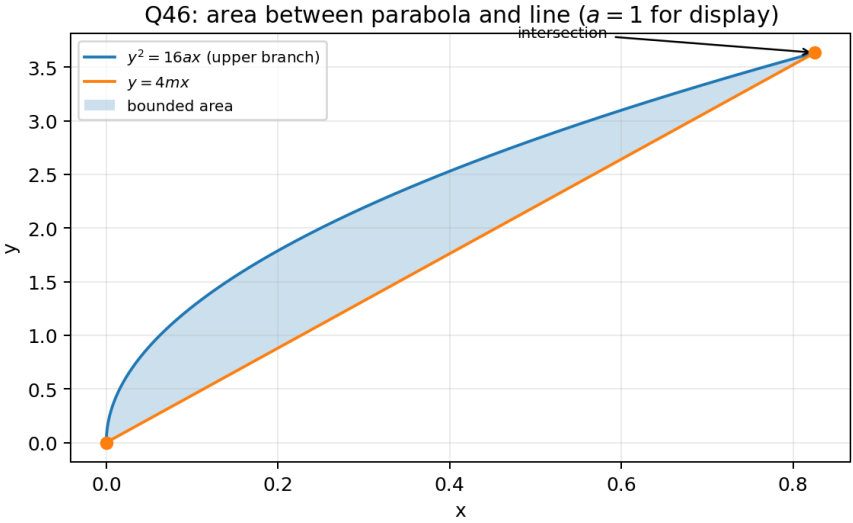
For positive slope, the area is the integral of the parabola minus the line:

$$\begin{aligned} A &= \int_0^{a/m^2} \left(4\sqrt{ax} - 4mx\right)\,dx \\ &= \left[\frac{8\sqrt{a}}{3}x^{3/2} - 2mx^2\right]_0^{a/m^2} \\ &= \frac{8a^2}{3m^3} - \frac{2a^2}{m^3} = \frac{2a^2}{3m^3}. \end{aligned}$$

Set the result equal to the given area:

$$\frac{2a^2}{3m^3} = \frac{a^2}{2} \quad\Longrightarrow\quad 4 = 3m^3 \quad\Longrightarrow\quad m = \sqrt[3]{\frac{4}{3}}.$$

The graph uses $a = 1$ for visibility; the value of m is independent of that scale.



Code reference – plotting

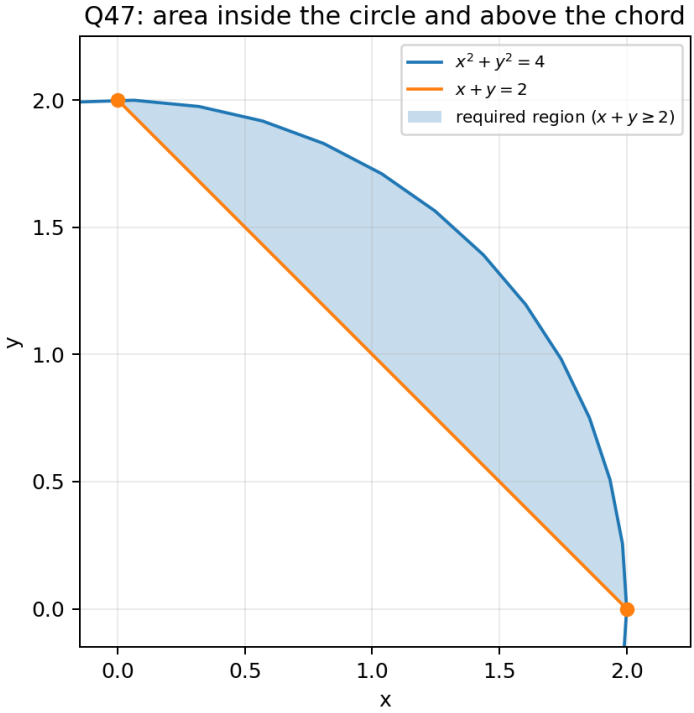
`codes/Q46.py`

1.2.47 Find the area of $\{(x, y) : x^2 + y^2 \le 4,\ x + y \ge 2\}$ by integration.

Solution. The chord $x + y = 2$ meets the circle at $(0, 2)$ and $(2, 0)$. For $0 \le x \le 2$, the upper circle boundary is $y = \sqrt{4 - x^2}$ and the line is $y = 2 - x$. Therefore

$$\begin{aligned} A &= \int_0^2 \left(\sqrt{4 - x^2} - (2 - x) \right) dx \\ &= \underbrace{\int_0^2 \sqrt{4 - x^2}\, dx}_{\text{quarter-circle area } = \pi} - \underbrace{\int_0^2 (2 - x)\, dx}_{\text{triangle area } = 2} \\ &= \pi - 2 \quad \text{square units.} \end{aligned}$$

The graph shades the circular segment above the chord.



Code reference – plotting

`codes/Q47.py`

Functions and relations:

1.2.48 Discuss continuity and differentiability of $f(x) = |x| + |x - 1|$ on $(-1, 2)$.

Solution. First split the absolute values at $x = 0$ and $x = 1$:

$$f(x) = \begin{cases} 1 - 2x, & -1 < x < 0, \\ 1, & 0 \leq x < 1, \\ 2x - 1, & 1 \leq x < 2. \end{cases}$$

Continuity. At $x = 0$, the left limit, right limit and function value are all 1. The same is true at $x = 1$. Each branch is continuous on its interval; hence f is continuous throughout $(-1, 2)$.

Differentiability at $x = 0$. The left-hand slope is -2 and the right-hand slope is 0:

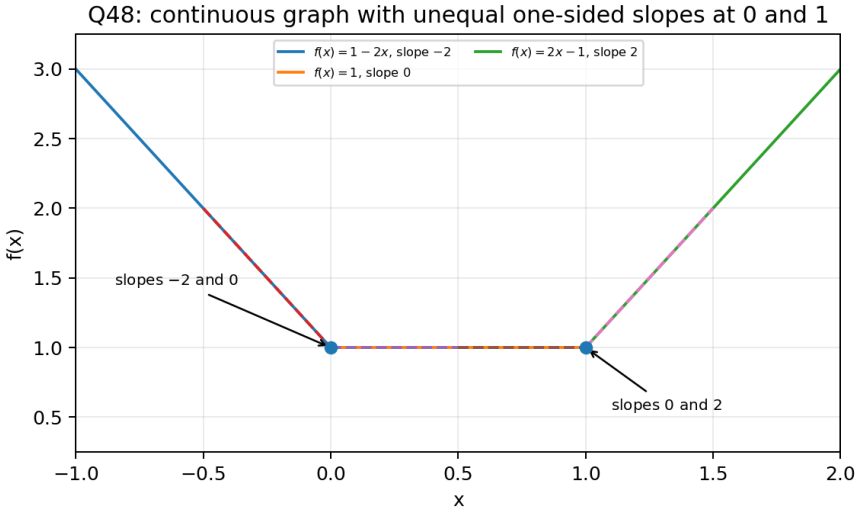
$$f'_-(0) = -2 \neq 0 = f'_+(0).$$

Thus f is not differentiable at 0.

Differentiability at $x = 1$. The left-hand slope is 0 and the right-hand slope is 2:

$$f'_-(1) = 0 \neq 2 = f'_+(1).$$

Thus f is not differentiable at 1. At other points the function is linear and differentiable. The plot shows these one-sided slopes: differentiability requires equal one-sided slopes, and the two corners fail that test.



Code reference – plotting and slope verification

`codes/Q48.py`

1.2.49 On $\mathbb{R}$, define $(a, b) \in R$ when $a - b + \sqrt{3}$ is irrational. Determine whether R is reflexive, symmetric and transitive.

Solution. We test each property separately.

1. Reflexive. Reflexivity requires $(a, a) \in R$ for every real a. Substituting $b = a$ gives

$$a - a + \sqrt{3} = \sqrt{3},$$

which is irrational. Therefore R is reflexive.

2. Symmetric. Symmetry requires $(a, b) \in R$ to imply $(b, a) \in R$. Choose $a = \sqrt{3}$ and $b = 0$. Then

$$a - b + \sqrt{3} = 2\sqrt{3}$$

is irrational, so $(a, b) \in R$. Reversing the pair gives

$$b - a + \sqrt{3} = 0,$$

which is rational, so $(b, a) \notin R$. Hence R is not symmetric.

3. Transitive. Transitivity requires $(a, b), (b, c) \in R$ to imply $(a, c) \in R$. Choose $a = \sqrt{3}$, $b = 0$, and $c = 2\sqrt{3}$. Then

$$a - b + \sqrt{3} = 2\sqrt{3}, \qquad b - c + \sqrt{3} = -\sqrt{3},$$

both irrational, so $(a,b),(b,c)\in R$. However,

$$a-c+\sqrt{3}=\sqrt{3}-2\sqrt{3}+\sqrt{3}=0,$$

which is rational. Thus $(a,c)\notin R$, and R is not transitive.
Conclusion. R is reflexive, but neither symmetric nor transitive.

1.2.50 On $\mathbb{R}\times\mathbb{R}$, define $(a,b)*(c,d)=(a+c,b+d)$. Prove commutativity and associativity, and find the identity and the inverse of $(3,-5)$.

Solution. Addition of real components is commutative and associative, so $*$ has both properties. Its identity is $(0,0)$, since $(a,b)*(0,0)=(a,b)$. The inverse of $(3,-5)$ is $(-3,5)$.

1.2.51 On $\{0,1,2,3,4,5,6\}$ define $a*b=a+b$ if $a+b<7$, and $a*b=a+b-7$ otherwise. Give the operation table and show that 0 is the identity and $7-a$ is the inverse for nonzero a.

Solution. This is addition modulo 7. Its table is

$*$	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

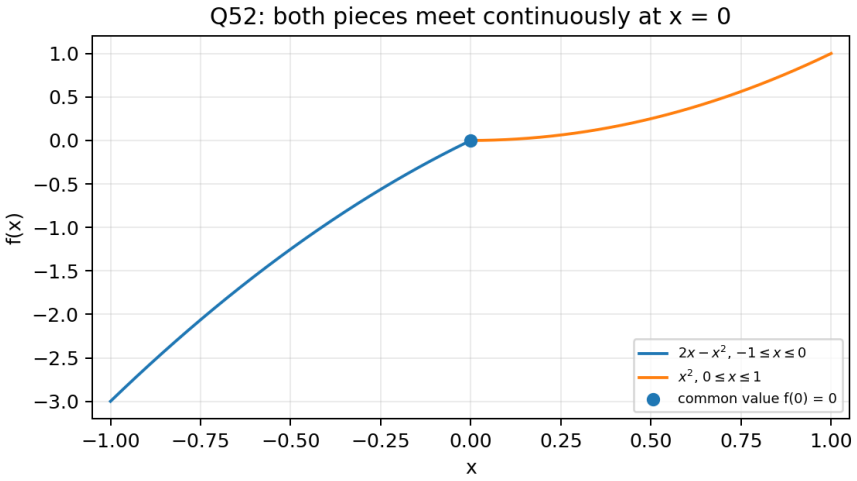
Row and column 0 reproduce every element, hence 0 is the identity. For each $a=1,\dots,6$, $a+(7-a)=7\equiv 0 \pmod 7$, so $7-a$ is its inverse.

1.2.52 Find any point of discontinuity of $f(x)=x-|x-x^2|$, $x\in[-1,1]$.

Solution. On $[-1,1]$, $x-x^2=x(1-x)$, whose sign changes only at 0. Thus

$$f(x)=\begin{cases}2x-x^2, & -1\le x\le 0,\\ x^2, & 0\le x\le 1.\end{cases}$$

Both pieces take value 0 at $x=0$ and meet continuously. The endpoints are also included continuously. Therefore there are no points of discontinuity.



Code reference – plotting

codes/Q52.py

Probability:

1.2.53 A man steps forward with probability 0.4 and backward with probability 0.6. Find the probability that after 5 steps he is one step away from his starting point.

Solution. Let k be the number of forward steps. The net displacement after five steps is

$$D = k - (5 - k) = 2k - 5.$$

One step away means $D = +1$ or $D = -1$. These occur when $k = 3$ or $k = 2$, respectively.

Step 1: Net displacement $+1$. Three steps are forward and two backward, so the binomial formula gives

$$\begin{aligned} P(D = 1) &= \binom{5}{3}(0.4)^3(0.6)^2 \\ &= 10(0.064)(0.36) = 0.2304. \end{aligned}$$

Step 2: Net displacement -1. Two steps are forward and three backward:

$$\begin{aligned} P(D = -1) &= \binom{5}{2}(0.4)^2(0.6)^3 \\ &= 10(0.16)(0.216) = 0.3456. \end{aligned}$$

Step 3: Add the disjoint events.

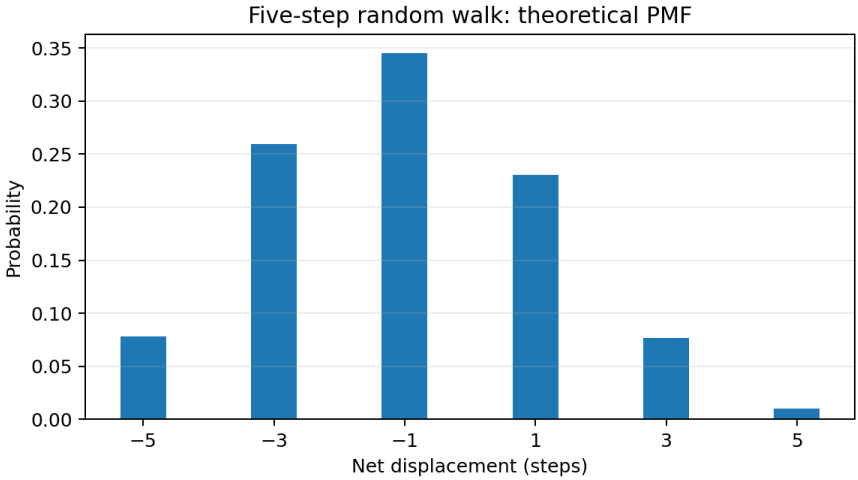
$$P(|D| = 1) = P(D = 1) + P(D = -1) = 0.2304 + 0.3456 = 0.576.$$

Generating function / Z-transform. For one step, the Laurent probability generating

function is $G(z) = 0.4z + 0.6z^{-1}$. For five independent steps,

$$G_5(z) = (0.4z + 0.6z^{-1})^5 = \sum_{k=0}^{5} \binom{5}{k} (0.4)^k (0.6)^{5-k} z^{2k-5}.$$

The coefficient of z^1 is 0.2304 and the coefficient of z^{-1} is 0.3456. Their sum is the required probability. The plot shows the theoretical PMF for every possible displacement.



Code reference – plotting and numerical verification

`codes/Q53.py`

1.2.54 A girl throws a die. If it shows 1 or 2, she tosses a coin three times; otherwise she tosses it once. Given that exactly one tail was obtained, find the probability that the die showed 3, 4, 5 or 6.

Solution. Let A mean that the die shows 1 or 2, B mean that it shows 3, 4, 5 or 6, and E mean that exactly one tail is observed.

Step 1: Find the prior probabilities.

$$P(A) = \frac{2}{6} = \frac{1}{3}, \qquad P(B) = \frac{4}{6} = \frac{2}{3}.$$

Step 2: Find the likelihood of E. If A occurs, three fair coin tosses are made. Exactly one tail can occur in $\binom{3}{1}$ of the 8 equally likely outcomes:

$$P(E \mid A) = \binom{3}{1} \left(\frac{1}{2}\right)^3 = \frac{3}{8}.$$

If B occurs, only one coin toss is made, so

$$P(E \mid B) = \frac{1}{2}.$$

Step 3: Calculate $P(E)$ by total probability.

$$\begin{aligned} P(E) &= P(E \mid A)P(A) + P(E \mid B)P(B) \\ &= \frac{3}{8} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{2}{3} \\ &= \frac{1}{8} + \frac{1}{3} = \frac{11}{24}. \end{aligned}$$

Step 4: Apply Bayes' theorem.

$$P(B \mid E) = \frac{P(E \mid B)P(B)}{P(E)} = \frac{(1/2)(2/3)}{11/24} = \frac{8}{11}.$$

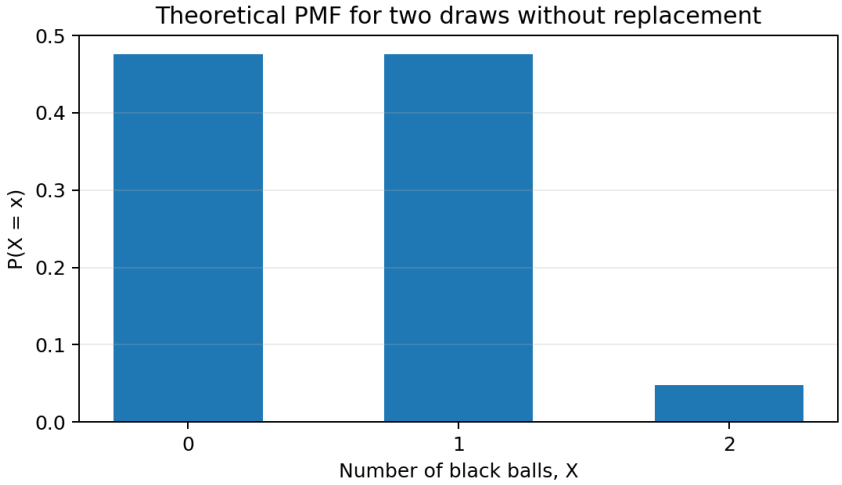
1.2.55 An urn contains 5 red and 2 black balls. Two balls are drawn without replacement. Let X be the number of black balls. State its possible values and find its mean and variance.

Solution. The possible values are $0, 1, 2$. The probabilities from the hypergeometric model are

$$P(X = 0) = \frac{10}{21}, \quad P(X = 1) = \frac{10}{21}, \quad P(X = 2) = \frac{1}{21}.$$

So $E[X] = 0(10/21) + 1(10/21) + 2(1/21) = 4/7$. Also $E[X^2] = 10/21 + 4/21 = 2/3$, hence

$$\mathrm{Var}(X) = E[X^2] - E[X]^2 = \frac{2}{3} - \frac{16}{49} = \frac{50}{147}.$$



Code reference – plotting and numerical verification

`codes/Q55.py`

1.2.56 In 3 trials of a binomial distribution, the probability of exactly 2 successes is 9 times the probability of 3 successes. Find the success probability.

Solution. Let the success probability be p and $q = 1 - p$. The probability-generating polynomial is

$$B(z) = (q + pz)^3 = q^3 + 3pq^2z + 3p^2qz^2 + p^3z^3.$$

The coefficient of z^2 is

$$P(X = 2) = \binom{3}{2} p^2 q = 3p^2 q,$$

and the coefficient of z^3 is

$$P(X = 3) = p^3.$$

Use $P(X = 2) = 9P(X = 3)$:

$$3p^2 q = 9p^3,$$
$$3p^2(1 - p) = 9p^3.$$

For the non-degenerate binomial case $p > 0$, divide by $3p^2$: $1 - p = 3p$. Thus $4p = 1$ and

$$p = \frac{1}{4}.$$

The polynomial's coefficients directly identify the two probabilities; no separate Z-transform is needed beyond this generating-polynomial viewpoint.

Code reference – numerical verification

codes/Q56.py

1.2.57 An urn contains 3 red and 5 black balls. A ball is drawn, its colour recorded and returned, and 2 more balls of that colour are added. Then two balls are drawn without replacement. Find the probability both are red.

Solution. If the first ball is red, with probability $3/8$, the urn becomes 5 red and 5 black; the chance of two reds is $(5/10)(4/9) = 2/9$. If the first ball is black, with probability $5/8$, the urn becomes 3 red and 7 black; the chance of two reds is $(3/10)(2/9) = 1/15$. Thus

$$P(\text{two red}) = \frac{3}{8}\frac{2}{9} + \frac{5}{8}\frac{1}{15} = \frac{1}{8}.$$

Code reference – numerical verification

codes/Q57.py

1.2.58 A man speaks truth 3 out of 5 times. He throws a die and reports that it is 4. Find the probability that it actually is 4.

Solution. Let A be the event that the die shows 4, and R the event that he reports 4. Assuming that when he lies each of the five wrong numbers is equally likely, $P(A) = 1/6$, $P(R|A) = 3/5$, and $P(R|A^c) = (2/5)(1/5) = 2/25$. Hence

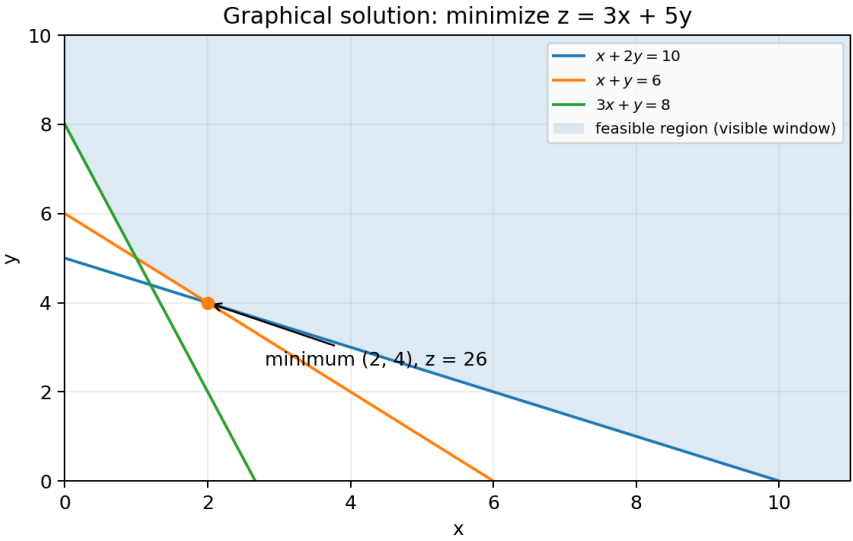
$$P(A|R) = \frac{(1/6)(3/5)}{(1/6)(3/5) + (5/6)(2/25)} = \frac{3}{5}.$$

Linear programming:

1.2.59 Solve graphically: minimise $z = 3x + 5y$ subject to $x + 2y \geq 10$, $x + y \geq 6$, $3x + y \geq 8$, $x, y \geq 0$.

Solution. The lower-boundary vertices are $(0, 8)$, $(1, 5)$, $(2, 4)$ and $(10, 0)$. Their objective values are respectively 40, 28, 26 and 30. The smallest value is attained at $(2, 4)$:

$$x = 2, \;\; y = 4, \;\; z_{\min} = 26.$$



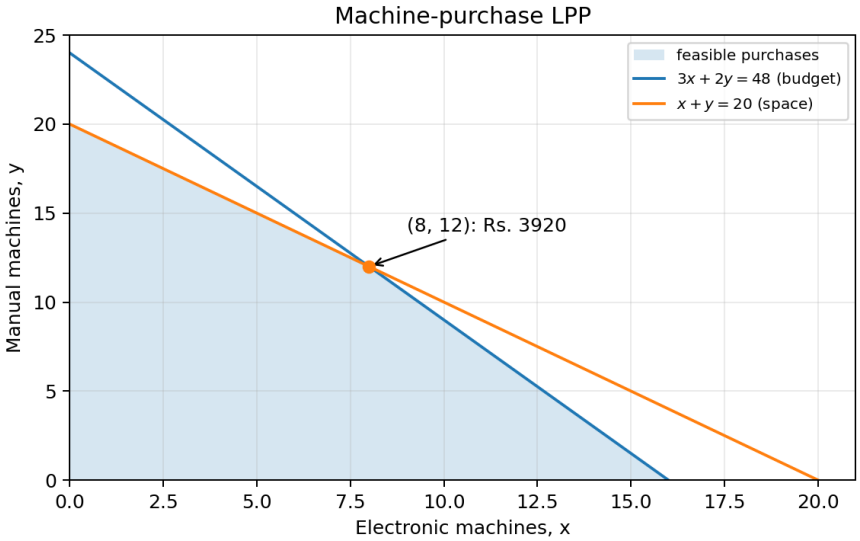
Code reference – plotting and numerical verification

`codes/Q59.py`

1.2.60 A dealer has Rs. 57,600 and space for at most 20 sewing machines. An electronic machine costs Rs. 3600 and earns Rs. 220 profit; a manual machine costs Rs. 2400 and earns Rs. 180 profit. How should he invest to maximise profit?

Solution. Let x and y be electronic and manual machines. The constraints are $3x + 2y \leq 48$, $x + y \leq 20$, $x, y \geq 0$, and profit is $P = 220x + 180y$. At the intersection of the two active limits, $3x + 2y = 48$ and $x + y = 20$, we get $(x, y) = (8, 12)$ and $P = 3920$. The axis vertices give profits Rs. 3520 and Rs. 3600. Therefore the best feasible integer plan is

$$8 \text{ electronic and } 12 \text{ manual machines; maximum profit Rs. } 3920.$$



Code reference – plotting and numerical verification

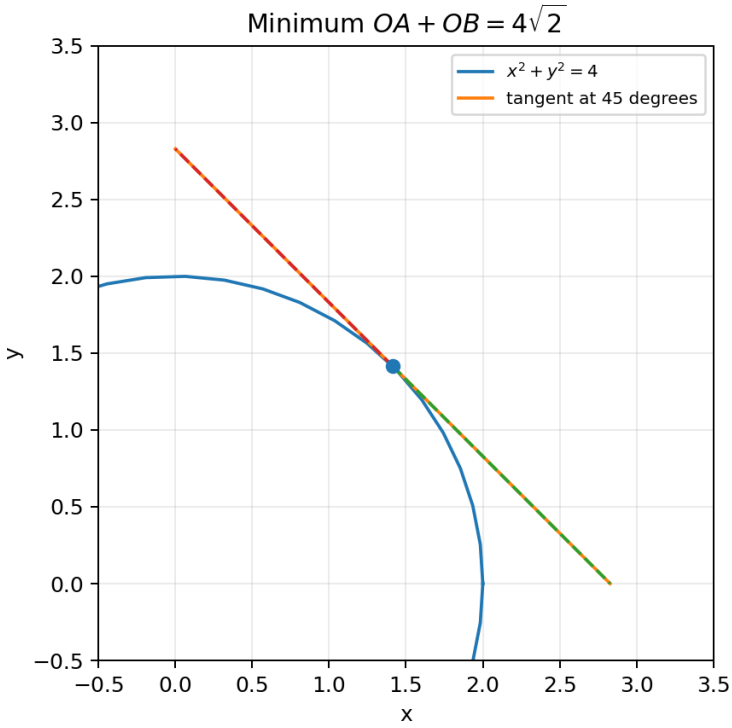
codes/Q60.py

Coordinate geometry and optimisation:

1.2.64 A tangent to $x^2 + y^2 = 4$ at a point in the first quadrant meets the positive axes at A and B. Find the minimum of $OA + OB$.

Solution. At the point $(2\cos\theta, 2\sin\theta)$, the tangent is $x\cos\theta + y\sin\theta = 2$. Thus $OA = 2\sec\theta$ and $OB = 2\csc\theta$. By symmetry (or differentiating), their sum is minimised when $\sin\theta = \cos\theta = 1/\sqrt{2}$. Hence

$$\min(OA + OB) = 2(\sqrt{2} + \sqrt{2}) = 4\sqrt{2}.$$



Code reference – plotting and numerical verification

`codes/Q64.py`

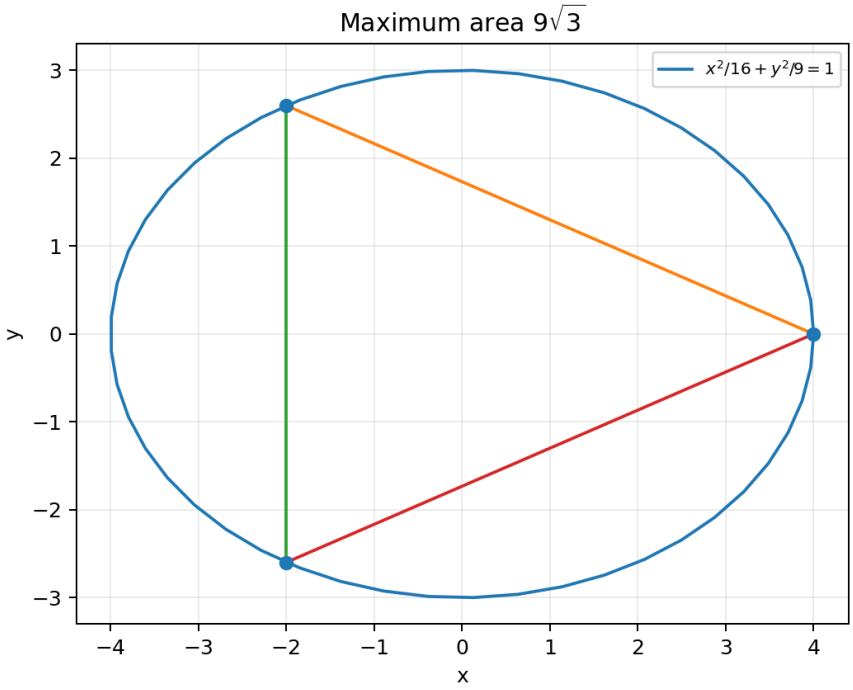
1.2.65 Find the maximum area of an isosceles triangle inscribed in $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with its vertex at an end of the major axis.

Solution. By symmetry, take the fixed vertex as $(4, 0)$ and the other two vertices as (u, v) and $(u, -v)$. From the ellipse, $v = 3\sqrt{1 - u^2/16}$. The area is

$$A(u) = v(4 - u) = 3(4 - u)\sqrt{1 - u^2/16}.$$

Differentiation gives $A'(u) = 0$ when $u^2 - 2u - 8 = 0$, i.e. $u = 4$ or $u = -2$. The endpoint $u = 4$ has zero area, so the maximum is at $u = -2$, where $v = 3\sqrt{3}/2$. Thus

$$A_{\max} = 6\left(\frac{3\sqrt{3}}{2}\right) = 9\sqrt{3}.$$



Code reference – plotting and numerical verification

codes/Q65.py